

Zenia Zuraïq

PH.D. STUDENT AND PRIME MINISTER'S RESEARCH FELLOW, INDIAN INSTITUTE OF SCIENCE

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Education

Indian Institute of Science (IISc)

Bengaluru, India

PH.D. THEORETICAL PHYSICS [EXPECTED 2027]

2022 - present

- Advisor: Prof. Banibrata Mukhopadhyay
- Thesis: Exploring unusual observations using massive, magnetized compact stars: theory and simulation

National Institute of Technology, Tiruchirappalli (NIT-T)

Tiruchirappalli, India

M.Sc. PHYSICS

2020-2022

- Advisor: Prof. Somnath Mukhopadhyay
- Thesis: Effects of fermion mass and strong two-body interactions on the mass-radius relationship of non-rotating compact stars

Stella Maris College, University of Madras

Chennai, India

B.Sc. PHYSICS

2017-2020

- Dissertation: Investigation on the optical properties of titanium nanoparticles in the treatment of waste water

Professional Experience

2022-present **Prime Minister's Research Fellow (PMRF)**, Indian Institute of Science

2023-present **Associate Member**, Indian Pulsar Timing Array (InPTA)

Publications

* - alphabetic ordering of authors

Major contributions

Mukhopadhyay, B., Roy, N., **Zuraïq, Z.**, Chakraborty, C., Rao, A. R. Exploring PSR J0901-4046 as an ultra-slowly rotating neutron star versus a white dwarf. *Submitted to Astrophys. J.* (**2026**)

Zuraïq, Z., Mukhopadhyay, B. Exploring the origin of slow-rotating pulsars: Emission from STARS simulated white dwarfs. *Submitted to Astrophys. J.* (**2026**)

Zuraïq, Z., Mukhopadhyay, B., Kumar, A., Sarkar, A., Hackett, A. J., Banerjee, P., Tout, C. A. Evolution of magnetized main-sequence stars to super-Chandrasekhar white dwarfs by STARS simulation. *Submitted to Astrophys. J. Lett.* (**2026**)

Zuraïq, Z., Susobhanan, A. Improving Pulsar Timing Array sensitivity using VLBI Astrometry. *Submitted to Phys. Rev. Lett.* (**2026**)

Zuraïq, Z., Mukhopadhyay, B. Anisotropic hybrid stars: Interplay of superconductivity and magnetic field leading to gravitational waves. arXiv: [2604.06308](https://arxiv.org/abs/2604.06308). *Submitted to JHEAp* (**2026**)

Zuraïq, Z., Das, M., Deb, D., Kalita, S., Weber, F., Mukhopadhyay, B. Anisotropic Compact Stars: Theory and Simulation from Microphysical Models to Macroscopic Structure and Observables. *Universe* 12 (**2026**) 5, 130.

DOI: <https://doi.org/10.3390/universe12050130>

Bhutani, A., Raj, N., **Zuraïq, Z.*** Dark, deep, deconfining: Phase transitions in neutron stars as powerful probes of hidden sectors. *Phys. Rev. D.* 113 (**2026**) 10, 103046. DOI: <https://doi.org/10.1103/yblk-411t>, arXiv: [2507.08076](https://arxiv.org/abs/2507.08076)

Zuraïq, Z., Mukhopadhyay, B., Weber, F. Massive neutron stars as mass gap candidates: Exploring equation of state and magnetic field. *Phys. Rev. D.* 109 (**2024**) 2, 023027. DOI: <https://doi.org/10.1103/PhysRevD.109.023027>, arXiv: [2311.02169](https://arxiv.org/abs/2311.02169)

Conference proceedings and white papers

Adhikari, P. *et al.* (61 authors including **Zurairq, Z.**). Strongly interacting matter in extreme magnetic fields. *Prog. Part. Nucl. Phys.* 146 (2026) 104199. DOI: <https://doi.org/10.1016/j.pnpnp.2025.104199>, arXiv: 2412.18632

Zurairq, Z., Kumar, A., Hackett, A. J., Mukhopadhyay, B. Simulating super-Chandrasekhar white dwarfs. *World Scientific* (2026). DOI: <https://doi.org/10.1142/14814> | *Proceedings of the Seventeenth Marcel Grossmann Meeting*

Zurairq, Z., Kumar, A., Hackett, A. J., Bhattarai, S., Tout, C. A., Mukhopadhyay, B. Simulating super-Chandrasekhar white dwarfs. *Astrophys. Space Sci. Proc.* 61 (2025) 397-407.

DOI: https://doi.org/10.1007/978-3-031-90186-7_32, arXiv: 2411.18692 | *Book chapter*

Zurairq, Z., Mukhopadhyay, B., Weber, F. Exploring massive neutron stars towards the mass gap: Constraining the high-density nuclear equation of state. *Astron. Rep.* 67 (2023) Suppl 2, S199-S206.

DOI: <https://doi.org/10.1134/S1063772923140214> | *Proceedings of the Fifth Zeldovich Meeting*

As part of InPTA collaboration

Tahbilda, H. *et al.* (31 authors including **Zurairq, Z.**). The Indian Pulsar Timing Array Data Release 2: III. Search for a Stochastic Gravitational Wave Background. arXiv: 2608.02808. *Submitted to Phys. Rev. D.* (2026)

Chowdhury, S. *et al.* (29 authors including **Zurairq, Z.**). Profile Reconstruction from Temporally Stable Emission Components for Timing PSR J1713+0747. arXiv: 2608.04108. *Submitted to JHEAp* (2026)

Nobleson, K. *et al.* (36 authors including **Zurairq, Z.**). The Indian Pulsar Timing Array Data Release 2: II. Customised single-pulsar noise analysis and noise budget. *JHEAp* 53 (2026) 100594.

DOI: <https://doi.org/10.1016/j.jheap.2026.100594>, arXiv: 2512.20455

Susobhanan, A. *et al.* (31 authors including **Zurairq, Z.**). Revisiting wideband pulsar timing measurements. *Mon. Not. Roy. Astron. Soc.* 547 (2026) 4, stag444. DOI: <https://doi.org/10.1093/mnras/stag444>, arXiv: 2512.01288

Chowdhury, S. *et al.* (30 authors including **Zurairq, Z.**). Effects of coronal mass ejection on PSR J1022+1001 and possible mode change of PSR J2145 - 0750 in the InPTA DR2. *JHEAp* 51 (2026) 100547.

DOI: <https://doi.org/10.1016/j.jheap.2025.100547>, arXiv: 2510.26594

Rana, P. *et al.* (47 authors including **Zurairq, Z.**). The Indian Pulsar Timing Array Data Release 2: I. Dataset and Timing Analysis. *Publ. Astron. Soc. Aust.* 42 (2025) p.e108, DOI: <https://doi.org/10.1017/pasa.2025.10066>, arXiv: 2506.16769

Awards, Fellowships, & Grants

2022-2027	Prime Minister's Research Fellowship, Government of India
2022	All India Rank 36 (out of 28135 candidates), awarded Junior Research Fellowship, National Eligibility Test, Council of Scientific and Industrial Research (CSIR-NET), India
2022	All India Rank 51 (out of 19375 candidates), Graduate Aptitude Test in Engineering (GATE)
2022	All India Rank 198 (96.12 Percentile), Joint Entrance Screening Test (JEST)
2022	Gold Medalist and Alumni Award for Outstanding Student, M.Sc. Physics, NIT-T
2020	All India Rank 669 (out of 16379 candidates), Joint Admission Test for Masters (JAM)
2020	Silver Medalist and Outgoing student of the year, B.Sc. Physics, Stella Maris College

Presentations

TALKS

Anisotropic hybrid stars: Interplay of superconductivity and magnetic field leading to gravitational waves
6th Zeldovich Meeting, Pescara, Italy | 07/26

Simulating magnetized white dwarfs: Chandrasekhar limit and beyond
Workshop: Magnetism in low-mass stars and their compact remnants, IIT Bombay, Mumbai, India | 12/25

Anisotropic neutron stars as mass gap candidates
Brainstorming workshop on neutron stars, IMSc, Chennai, India | 08/25

Explaining unusual observations using magnetized white dwarfs
International Pulsar Timing Array (IPTA) Meeting 2025, California Institute of Technology, Pasadena, USA | 06/25

Addressing fundamental puzzles with the neutron star EOS
University of California, Riverside, USA | 06/25

Simulating magnetized white dwarfs: Chandrasekhar limit and beyond
GW Paleontology group meeting, University of California San Diego, USA | 06/25

Explaining unusual observations using magnetized white dwarfs
43rd Meeting of the Astronomical Society of India, NIT Rourkela, Odisha, India | 02/25

Simulating magnetized white dwarfs by time evolution: Chandrasekhar limit and beyond **[invited]**
4th Symposium of the BRICS Association on Gravity, Astrophysics and Cosmology, SGT University, Gurugram, India | 12/24

Anisotropic neutron stars as mass gap candidates
Conference on Classical and Quantum Gravity, CUSAT, Cochin, India | 11/24

A journey towards compact star physics... and beyond! **[invited]**
Research Day talk, Stella Maris College, Chennai, India | 09/24

Massive, magnetized neutron stars as mass gap objects
Parallel session: Galactic and extragalactic magnetars: recent observations and theoretical progress
17th Marcel Grossmann meeting, Pescara, Italy | 07/24

Simulating magnetized white dwarfs by time evolution: Chandrasekhar limit and beyond
Parallel session: Massive white dwarfs and related phenomena
17th Marcel Grossmann meeting, Pescara, Italy | 07/24

Simulating super-Chandrasekhar white dwarfs
International Symposium on Recent Development in Relativistic Astrophysics, SRM Sikkim, Gangtok, India | 12/23

Massive, magnetized compact stars: Theory and Simulation
10th International Conference on Gravitation and Cosmology, IIT Guwahati, Assam, India | 12/23

Massive neutron stars: Effect of EOS vs magnetic field **[invited]**
Online: London-Oldenburg Relativity Seminar | 10/23

Massive neutron stars: effects of EOSs and magnetic field
Online: Strongly interacting matter in extreme magnetic fields, ECT* workshop | 09/23

Exploring massive neutron stars towards the mass gap: Constraining the high density nuclear equation of state
5th Zeldovich Meeting, Yerevan, Armenia | 06/23

POSTERS

Simulating magnetized white dwarfs by time evolution: Chandrasekhar limit and beyond
24th European Workshop on White Dwarfs, ISTA Klosterneuburg, Austria | 08/26

Simulating magnetized white dwarfs by time evolution: Chandrasekhar limit and beyond
In-House Symposium 2025, Department of Physics, Indian Institute of Science, Bengaluru, India | 11/25

Explaining unusual observations using magnetized white dwarfs
In-House Symposium 2024, Department of Physics, Indian Institute of Science, Bengaluru, India | 11/24

Massive neutron stars as mass gap candidates **[invited]**
PMRF Annual Symposium 2024, IIT Indore, Indore, India | 03/24

Simulating super-Chandrasekhar white dwarf from main sequence star: Exploring stellar evolution codes STARS and MESA
Parallel session: Stars, Interstellar Medium, and Astrochemistry in Milky Way
42nd annual meeting of the Astronomical Society of India, Indian Institute of Science, Bengaluru, India | 02/24

Massive neutron stars as mass gap candidates **[Best poster award]**
In-House Symposium, Department of Physics, Indian Institute of Science, Bengaluru, India | 11/23

Simulating magnetised super-Chandrasekhar white dwarfs using STARS
3rd Conference on Plasma Simulations, Raman Science Center, IIA, Leh, India | 07/23

Conferences & other visits

CONFERENCES ATTENDED

- 24th European Workshop on White Dwarfs, ISTA Klosterneuburg, Austria | 08/26
- 6th Zeldovich Meeting, Pescara, Italy | 07/26
- Workshop: Magnetism in low-mass stars and their compact remnants, IIT Bombay, Mumbai, India | 12/25
- Brainstorming workshop on neutron stars, IMSc, Chennai, India | 08/25
- International Pulsar Timing Array (IPTA) Meeting 2025, California Institute of Technology, Pasadena, USA | 06/25
- 43rd annual meeting of the Astronomical Society of India (ASI), NIT Rourkela, Rourkela, India | 02/25
- 4th Symposium of the BRICS Association on Gravity, Astrophysics and Cosmology, SGT University, Gurgaon, India | 12/24
- Conference on Classical and Quantum Gravity, CUSAT, Cochin, India | 11/24
- 17th Marcel Grossmann meeting, Pescara, Italy | 07/24
- PMRF Annual Symposium 2024, IIT Indore, Indore, India | 03/24
- Meeting on Pulsar Timing Array Experiments: Present and Future of Indian Collaboration, IMSc, Chennai, India | 02/24
- 42nd annual meeting of the Astronomical Society of India (ASI), Indian Institute of Science, Bengaluru, India | 02/24
- International Symposium on Recent Developments in Relativistic Astrophysics, SRM Sikkim, Gangtok, India | 12/23
- 10th International Conference on Gravitation and Cosmology (ICGC), IIT Guwahati, Guwahati, India | 12/23
- 3rd Conference on Plasma Simulations, Raman Science Center, IIA, Leh, India | 07/23
- 5th Zeldovich meeting, Yerevan, Armenia | 06/23

RESEARCH VISITS

- University of Warsaw | 07/26, 07/24
- ICRA Net-Ferrara, University of Ferrara | 07/26, 06/24
- University of California, Riverside | 06/25
- University of California, San Diego | 06/25
- Department of Physics, San Diego State University | 06/25
- Instytut Fizyki Teoretycznej, Uniwersytet Wrocławski (University of Wrocław) | 07/24

INTERNSHIPS AND SUMMER SCHOOLS

- IIA Summer Programme, IIA, Bengaluru, India | 07/21
- Vigyan Vidushi, TIFR, Mumbai, India | 05-06/2021
- Introductory Summer School in Astronomy and Astrophysics, IUCAA, Pune, India | 05/21
- National Institute on Undergraduate Sciences (NIUS) Camp (15.1), TIFR, Mumbai, India | 06/18
- Summer Intern, IIUCNN, Mahatma Gandhi University, Kottayam, India | 05/18

Teaching Experience

08/26 - present	Mechanics and Waves II , Teaching Assistant Undergraduate course at Azim Premji University, Bengaluru, India
01-04/26	Particle Physics and the Standard Model , PMRF Teaching Assistant NPTEL online course Link
08-12/25	Quantum Mechanics I , Teaching Assistant Undergraduate course at IISc
08-10/25	Mechanics and Waves II , Teaching Assistant Undergraduate course at Azim Premji University, Bengaluru, India
07-10/25	The Joy of Computing using Python , PMRF Teaching Assistant NPTEL online course Link
01-04/25	Intermediate Mechanics (Lab) , Teaching Assistant Undergraduate course at IISc
01-03/25	Programming, Data Structures And Algorithms Using Python , PMRF Teaching Assistant NPTEL online course Link
07-10/24	The Joy of Computing using Python , PMRF Teaching Assistant NPTEL online course Link
04-06/24	Introductory Astronomy and Astrophysics , Instructor, value added course at M.S. Ramaiah University of Applied Sciences, Bengaluru, India Link
01-03/24	Nuclear Astrophysics , PMRF Teaching Assistant NPTEL online course Link
08-12/23	Introductory Mechanics , Teaching Assistant Undergraduate course at IISc [<i>Best TA award</i>]
08-12/23	Introductory Astronomy , Instructor to Classes VI-X, SPARC Public School, Bengaluru, India
01-06/23	High school physics , PMRF Teaching Assistant for Class XI, Kendriya Vidyalaya, IISc Bengaluru, India

Mentoring

2023-2024	Surendra Bhattarai , Masters thesis student, IISER Kolkata, India [currently Ph.D. student at Yale University]
2023	Ashmita Acharya , Undergraduate student, M.S. Ramaiah University of Applied Sciences, Bengaluru, India [currently masters student at Uppsala University]

Skills

- **Programming Languages:** Python, C++, MATLAB, Mathematica
- **Astrophysics Codes:** Cambridge STARS, MESA, PSRCHIVE, Tempo2, ENTERPRISE, PINT
- **Tools:** LaTeX, MS Office
- **OS:** Linux, Windows, macOS

Service & Outreach

Peer review: Reviewer for European Physical Journal C, Astrophysics and Space Science Proceedings

Public talk: *The Life and Death of Stars: Stellar Physics and Evolution*
IISc PRAVEGA Astrophysics Workshop, Bengaluru, India | 01/26, 05/26

Volunteer: IISc Open Day

Public Talk: *The Sun and our Solar System* (2023), Conducted quizzes (2024, 2026)

Public talk [**invited**]: *Looking out at the Universe in Radio Waves: From Communication to Astronomy*
IEEE ComSoc SCT SBC: ComConnect 1.0, SCT College of Engineering, Thiruvananthapuram, India | 09/25

Outreach talk: *Looking out at the universe: multi-wavelength astronomy*
SERB outreach program for class X students, IISc, Bengaluru, India | 10/24

Volunteer: 42nd annual meeting of the Astronomical Society of India, IISc, Bengaluru, India | 02/24

External expert: *National Workshop on physical sciences for CSIR-UGC NET and GATE aspirants*, NIT-T, India | 05/23

Outreach talks: *Overview of fermionic compact stars*
Talk I: Non-interacting stars and white dwarfs, Talk II: Introducing interactions
PMRF outreach talk series: Christ University, Bengaluru, India | 02-04/23

Talk: *Equation of motion of double pendulum using different mechanics and MATLAB*
NIT-T weekly colloquium | 06/21

Talk: *Exoplanet detection by transit method*
NIT-T weekly colloquium | 10/20

Public Talk: *Space Debris*
Loyola Astronomy and Rocketry Club, Chennai, India | 02/19

References

- **Prof. Banibrata Mukhopadhyay**, Department of Physics, Indian Institute of Science, India
bm@iisc.ac.in
- **Prof. Fridolin Weber**, Department of Physics, San Diego State University, USA
fweber@physics.ucsd.edu
- **Prof. Tomasz Bulik**, Director of the Astronomical Observatory, University of Warsaw, Poland
tb@astrouw.edu.pl
- **Prof. Jorge Rueda**, ICRANet, Dipartimento di Fisica e Scienze della Terra, Università di Ferrara, Italy
jorge.rueda@icra.it
- **Prof. Abhimanyu Susobhanan**, School of Physics, IISER Thiruvananthapuram, India
abhimanyu@iisertvm.ac.in
- **Prof. Nirmal Raj**, Centre for High Energy Physics, Indian Institute of Science, India
nraj@iisc.ac.in